

SUBHAJIT CHATTOPADHYAY

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Education

Indian Institute of Technology (IIT) Bhubaneswar	2024 - Present
Ph.D. Research Scholar, School of Mechanical Sciences	
National Institute of Technical Teachers Training & Research, Kolkata	2022 - 2024
M.Tech in Manufacturing Technology	CGPA: 8.24/10 (Distinction)
Meghnad Saha Institute of Technology, Kolkata	2019 -2022
B.Tech in Mechanical Engineering	CGPA: 8.25/10 (Distinction)

Research Interests

Lightweight Entropic Alloys, Magnesium-Based Alloys, Powder Metallurgy, Additive Manufacturing, Composite Materials, Alloy Development, Mechanical Behavior of Materials, High-Temperature Deformation, Microstructural Characterization, Surface Engineering

Research Experience

M.Tech Thesis	June 2023 - June 2024
<i>Fabrication and Characterization of Nano-SiC/Eggshell Reinforced Aluminium Alloy Hybrid Nanocomposites.</i>	
Supervisors: Prof. (Dr.) Debi Prasad Mishra, Dr. Subrata Mondal	
- Fabricated green hybrid aluminium matrix composites using powder metallurgy. - Conducted mechanical, compressive, corrosion, and microstructural analyses. - Studied grain morphology and reinforcement dispersion mechanisms.	
Course Project	Jan 2023 - June 2023
<i>Fabrication and Characterization of Aluminium Matrix Composite Reinforced with Aluminium Oxide (Al_2O_3).</i>	
Supervisor: Dr. Subrata Mondal	
- Developed Al-based composites via powder metallurgy route. - Evaluated hardness and microstructural evolution.	
B.Tech Final Year Project	Jan 2022 - June 2022
<i>Enhancing Machinability of Mild Steel Using Thermal In-Flux</i>	
Supervisors: Dr. Manish Mukhopadhyay, Debolina Dutta	
- Compared conventional and hot machining processes. - Demonstrated feasibility of hot machining for improved performance.	

Publications

- Chattopadhyay, S., Mahapatra, S. D., & Mandal, N. K. (2024). Advancements and challenges in additive manufacturing: A comprehensive review. *Engineering Research Express*, 6(1), 012505. <https://doi.org/10.1088/2631-8695/ad30b1>
- Chattopadhyay, S., Mondal, S., & Mishra, D. P. (2025). Morphology and properties of aluminium alloy matrix hybrid nanocomposites reinforced with nano-SiC and eggshell particles. *Journal of Materials Engineering and Performance* 1-16. <https://doi.org/10.1007/s11665-025-10813-6>

- Hazra, S., **Chattopadhyay, S.**, Chakravarty, S., & Mandal, N. K. (2024). Tribology of Mg-B₄C nanocomposites fabricated following powder metallurgy route. *Engineering Research Express*, 6(3), 035525.
<https://doi.org/10.1088/2631-8695/ad652f>
- Hazra, S., Ganguly, G., & **Chattopadhyay, S.** (2024). Slicing on the tessellated model for the 3D printing process based on variable thickness layers. *Journal of Manufacturing Engineering*, 19(3), 082–088.
<https://doi.org/10.37255/jme.v19i3pp082-088>

Technical Skills

Experimental: Tensile Testing, Vickers & Rockwell Hardness, Optical & SEM Analysis, Sample Preparation, Stir Casting, Laser Surface Texturing

Manufacturing: Lathe, Drill Press, Powder Metallurgy

Software: AutoCAD, SolidWorks, MATLAB, Python (Basic), Thermo Calc, MS Office

Achievements & Awards

- Swami Vivekananda Merit-cum-Means Scholarship (INR 60,000 per annum).
- Subject Matter Expert – Mechanical Engineering (Chegg India, Tutor Champs).
- AutoCAD & SolidWorks Certification (Internshala).
- Best Player – Subdivision Volleyball Tournament.

Selected as a Peer Reviewer for international journals:

- **Materials Today: Physics** (Elsevier)
- **Materials Research Express** (IOP Publishing)

Personal Details

Date of Birth: 02 August 1999 — Gender: Male — Nationality: Indian
Languages: English, Hindi, Bengali

Reference

Dr. Manish Mukhopadhyay

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Dr. Arpan Kumar Mondal

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